

Day 2 PM: quanteda

Follow along:

corpusmasterclass2.Rmd

quanteda: Quantitative Analysis of Textual Data

quanteda is an R package for managing and analyzing textual data developed by [Kenneth Benoit](#), [Kohei Watanabe](#), and other contributors. Its initial development was supported by the European Research Council grant ERC-2011-StG 283794-QUANTESS.

The package is designed for R users needing to apply natural language processing to texts, from documents to final analysis. Its capabilities match or exceed those provided in many end-user software applications, many of which are expensive and not open source. The package is therefore of great benefit to researchers, students, and other analysts with fewer financial resources. While using **quanteda** requires R programming knowledge, its API is designed to enable powerful, efficient analysis with a minimum of steps. By emphasizing consistent design, furthermore, **quanteda** lowers the barriers to learning and using NLP and quantitative text analysis even for proficient R programmers.

Links

Download from CRAN at
<https://cloud.r-project.org/package=quanteda>

Browse source code at
<https://github.com/quanteda/quanteda/>

Report a bug at
<https://github.com/quanteda/quanteda/issues>

License

[GPL-3](#)

Citation

LIWCing at SOTUs...





DISCOVER LIWC2015

LIWC2015 is the gold standard in computerized text analysis. Learn how the words we use in everyday language reveal our thoughts, feelings, personality, and motivations. Based on years of scientific research, LIWC2015 is more accurate, easier to use, and provides a broader range of social and psychological insights compared to earlier LIWC versions. Check it out.

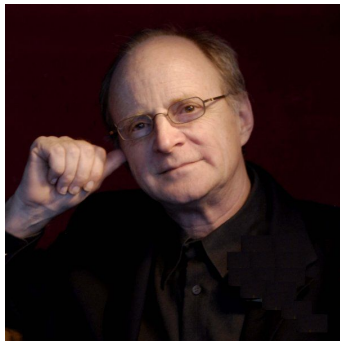
[BUY ACADEMIC VERSION](#)

[BUY COMMERCIAL VERSION & API](#)

INTRODUCING LIWC-22

A NEW SET OF TEXT ANALYSIS TOOLS AT YOUR FINGERTIPS

People reveal themselves by the words they use. Using LIWC-22 to analyze others' language can help you understand their thoughts, feelings, personality, and the ways they connect with others. It can give you insights you've never had before into the people and world around you.

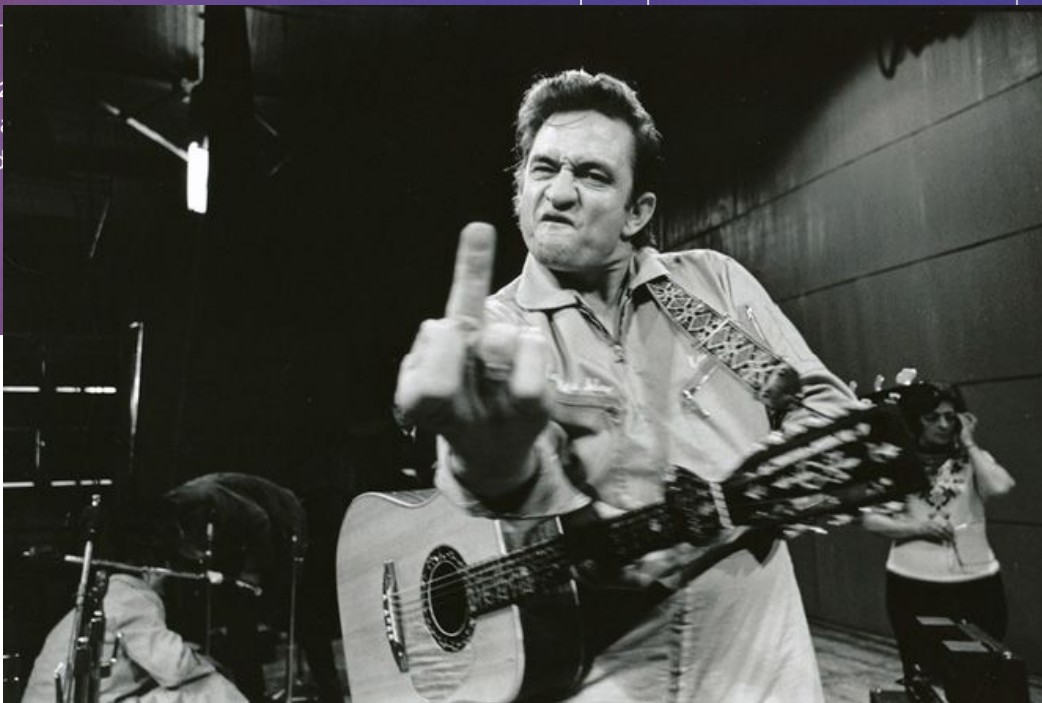
[BUY LIWC NOW](#)[CONTACT US](#)

INTRODUCING LIWC-22

A NEW SET OF TEXT ANALYSIS TOOLS AT YOUR FINGERTIPS

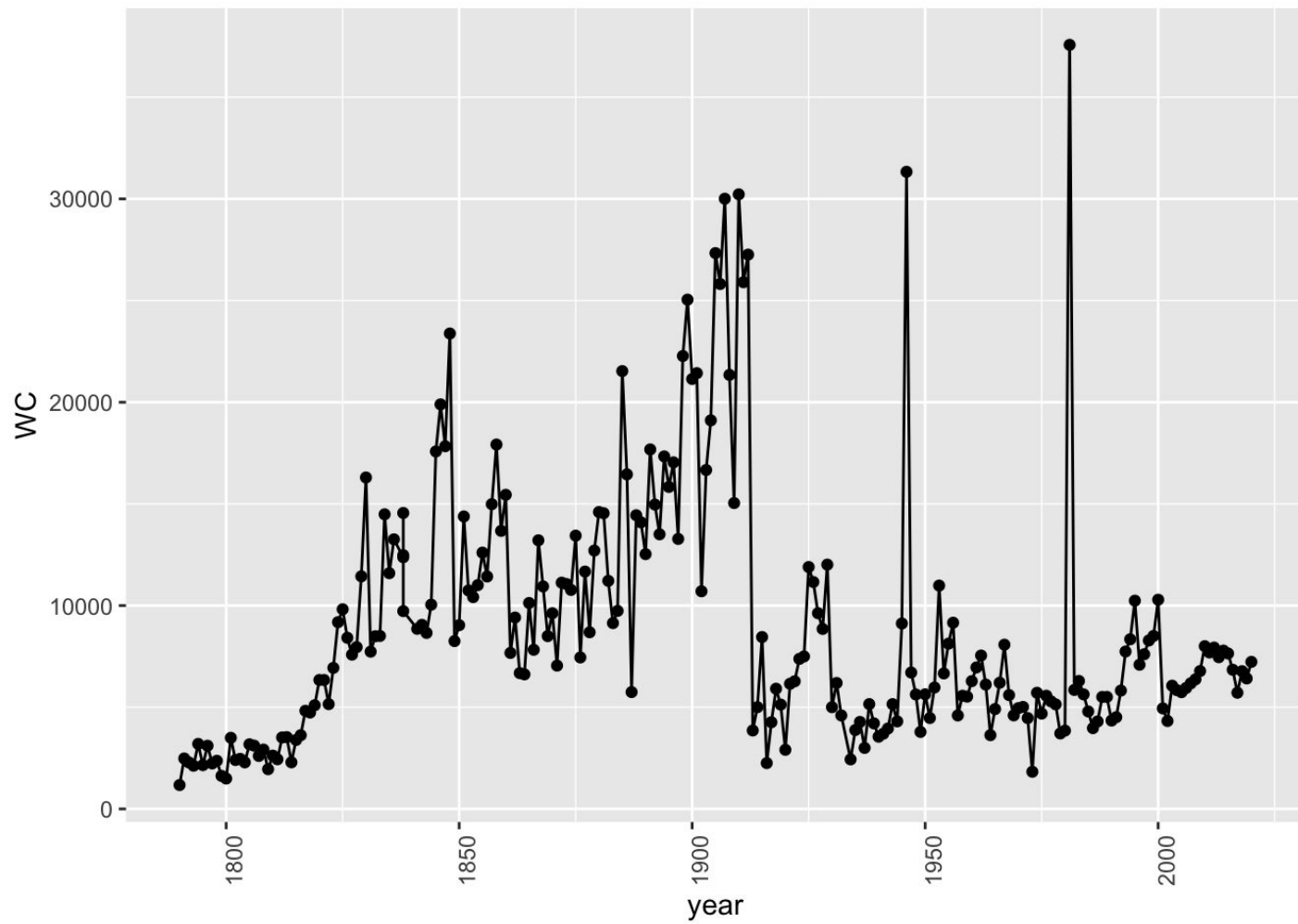
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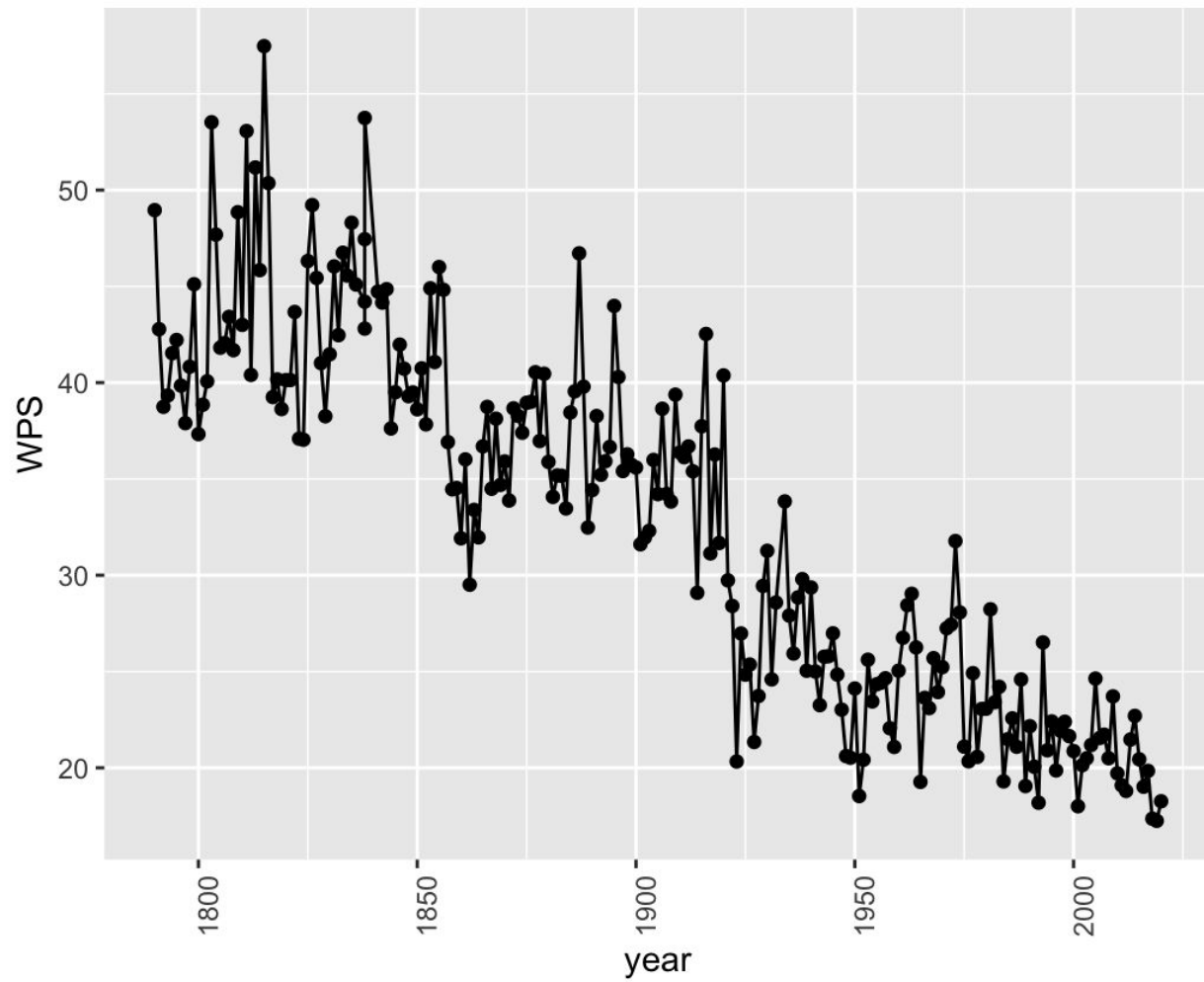
People reveal themselves by the words they use. Using LIWC-22, you understand their thoughts, feelings, personality, and the world they live in. You gain insights you've never had before into the people and world around you.

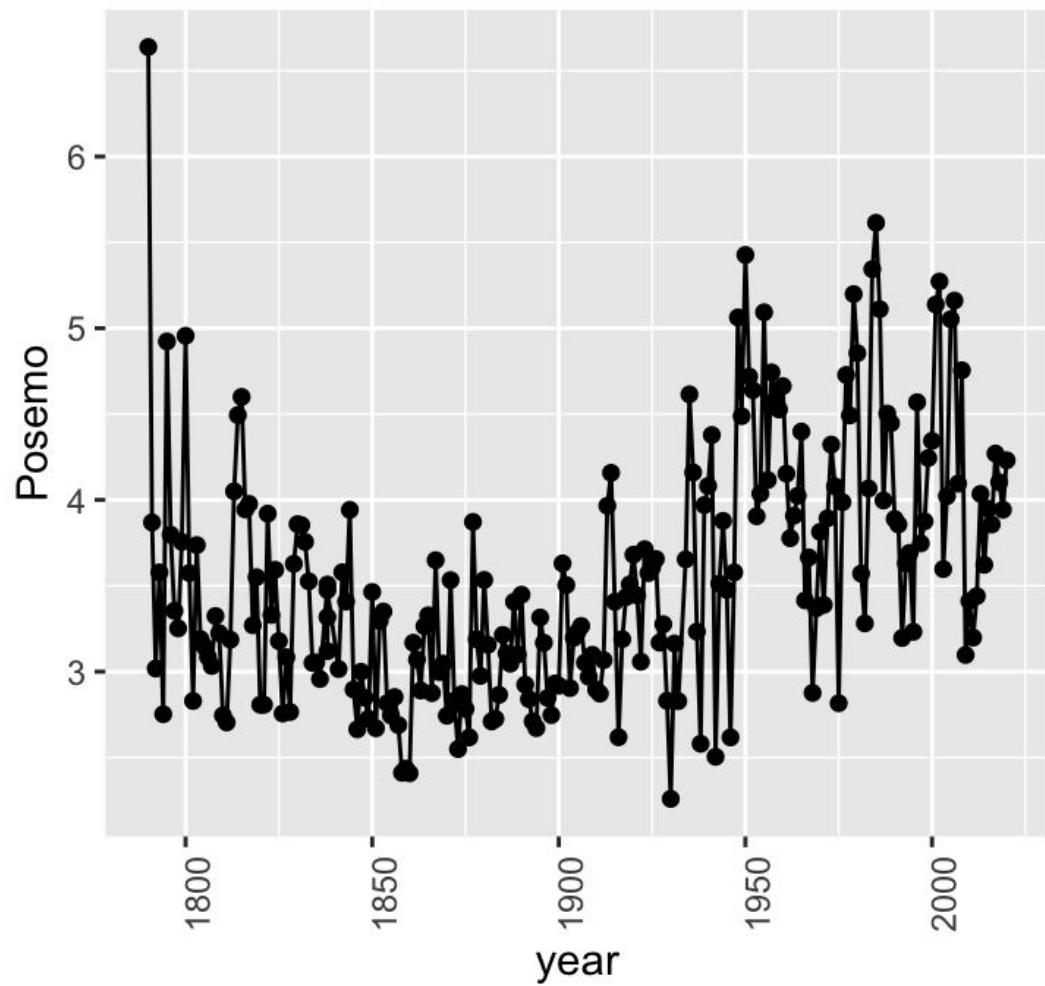


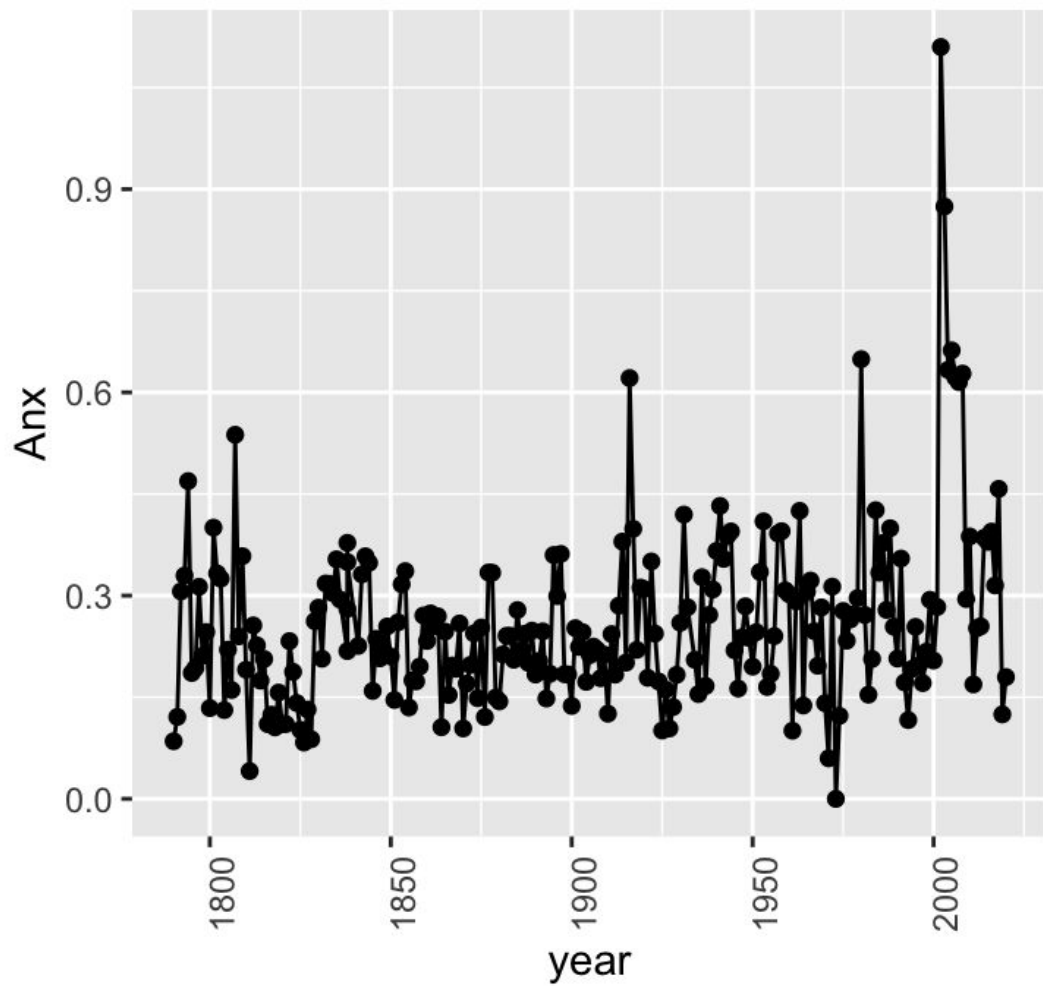
president ↕	year ↕	Segment ↕	WPS ↕	WC ↕	Sixltr ↕	Dic ↕	function ↕	pronoun ↕	apron ↕	i ↕	we ↕	you ↕	shehe ↕	they ↕	ipron ↕	article ↕	prep ↕
Washington	1790	1	48.95833	1175	26.72	230.89	51.83	9.70	5.19	1.28	1.11	1.70	0.00	1.11	4.51	10.47	18.21
Washington	NA	2	39.55263	1503	25.62	242.32	52.63	11.58	5.66	1.00	1.86	1.33	0.13	1.33	5.92	9.91	17.83
Washington	1791	3	42.77586	2481	26.72	219.51	51.43	7.54	2.90	0.36	0.24	1.05	0.00	1.25	4.64	11.89	17.98
Washington	1792	4	38.74576	2286	25.55	226.20	51.36	8.66	3.76	1.31	0.66	0.74	0.00	1.05	4.90	10.24	18.81
Washington	1793	5	39.33333	2124	23.49	230.51	52.97	8.47	4.14	1.04	1.13	0.52	0.14	1.32	4.33	10.83	17.98
Washington	1794	6	41.53247	3198	25.30	223.20	51.81	8.85	3.97	0.97	0.97	0.19	0.47	1.38	4.88	10.60	16.79
Washington	1795	7	42.21569	2153	27.73	230.70	50.30	9.43	4.97	1.07	2.60	0.42	0.23	0.65	4.46	9.89	16.44
Washington	1796	8	39.84615	3108	25.32	221.33	50.32	7.75	3.22	0.87	0.90	0.23	0.26	0.97	4.54	10.68	18.15
Adams	1797	9	37.89831	2236	25.22	222.45	50.04	7.92	4.16	0.94	1.21	0.36	0.13	1.52	3.76	9.35	17.89
Adams	1798	10	40.82759	2368	25.42	226.82	51.86	9.76	4.43	0.59	1.98	0.46	0.51	0.89	5.32	11.06	16.30
Adams	1799	11	45.11111	1624	26.79	218.97	50.80	9.05	4.43	1.23	1.35	0.43	0.43	0.99	4.62	10.96	17.98
Adams	1800	12	37.32500	1493	26.05	233.42	51.37	9.85	4.82	0.67	2.21	0.80	0.07	1.07	5.02	9.71	17.55
Jefferson	1801	13	38.85556	3497	24.96	247.35	52.04	10.64	5.32	0.60	2.20	0.66	0.14	1.72	5.32	8.06	17.53
Jefferson	1802	14	40.06667	2404	23.04	230.78	50.58	9.78	4.66	0.25	2.58	0.58	0.08	1.16	5.12	9.28	17.64

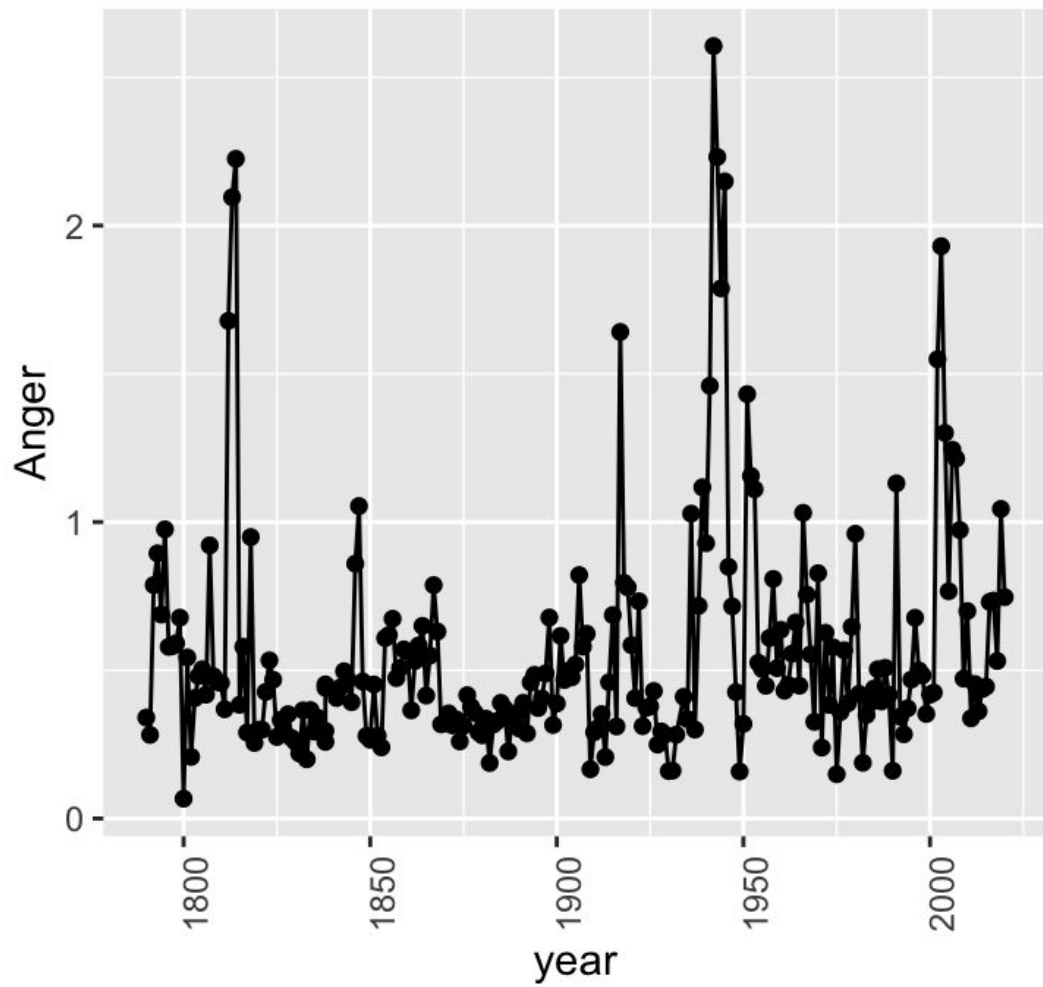
%		%		%		%	
1	Function	2	Family	1	nuke	1	admiration
2	Pronoun	2	Group	2	weapon	2	affect
3	Apron	3	Reciprocity	3	proliferation	3	anger
4	I	4	Heroism	4	ban	4	base
5	We	5	Deference	5	energy	5	chastity
6	You	6	Fairness	6	intermediate-range	6	comedy
7	SheHe	7	Property	7	reactor	7	conscience
8	They	%		8	arms	8	contempt
9	Ipron	a favor 3		9	nonproliferation	9	courage
10	Article	a favour 3		10	evaluation	10	cruelty
11	Prep	abase 5		11	iran	11	culture
12	Auxverb	abased 5		12	full-scope	12	curiosity
13	Adverb	abasedly 5		13	nassau	13	disgust
14	Conj	abasedness 5		14	reprocessing	14	doubt
15	Negate	abasement 5		15	explosions	15	drive
20	Verb	abasement's 5		16	regime	16	emotion
21	Adj	abasements 5		17	chemical	17	exemplar
22	Compare	abaser 5		18	suppliers	18	fear
23	Interrog	abaser's 5		19	strategic	19	forgetting
24	Number	abasers 5		20	infce	20	guilt
25	Quant	abases 5		21	tlatelolco	21	health
30	Affect	abid 5		22	stockpile	22	herd
31	Posemo	abjection 5		23	warheads	23	honesty
32	Negemo	abnegate 5		24	fuel	24	humility
33	Anx	abnegated 5		25	korea	25	humor
34	Anger	abnegates 5		26	spread	26	instinct
35	Sad	abnegating 5		27	missiles	27	integrity
40	Social	abnegation 5		28	biological	28	joy
41	Family	abnegation's 5		29	cycle	29	justice
42	Friend	abnegations 5		30	repository	30	laughter
43	Female	abnegative 5				31	life
44	Male	abnegator 5		%		32	mistrust
50	CogProc	abnegator's 5		atomic 1		33	modesty
51	Insight	abnegators 5		nuclear 1		34	nobility
52	Cause	absolve 3		nuke* 1		35	obligation
53	Discrep	accede 5		weapon* 2		36	pod
54	Tentat	accomplice 2		proliferat* 3		37	perspective
		accomplice's 2		ban 4		38	rank
		accomplices 2		bans 4		39	resentment
		accompliceship 2		banned 4		40	respons
		achiever 4		banning 4		41	revenge
		acolyte 5		energ* 5		42	sadness
		acquiesce 5		intermediate-range 6		43	shame
		acquiesced 5		reactor* 7		44	solitude
		acquiescement 5		arms 8		45	style
		acquiescence 5		non-proliferat* 9		46	surprise
		acquiescence's 5		nonproliferat* 9		47	tragedy
		acquiescences 5		evaluat* 10		48	trust
		acquiescency 5		iran* 11		49	type
		acquiescent 5		full-scope 12		50	value
		acquiescently 5		nassau 13		51	virtue
		acquiescents 5		reprocess* 14		52	vice
		acquiescer 5		explosi* 15		53	wto
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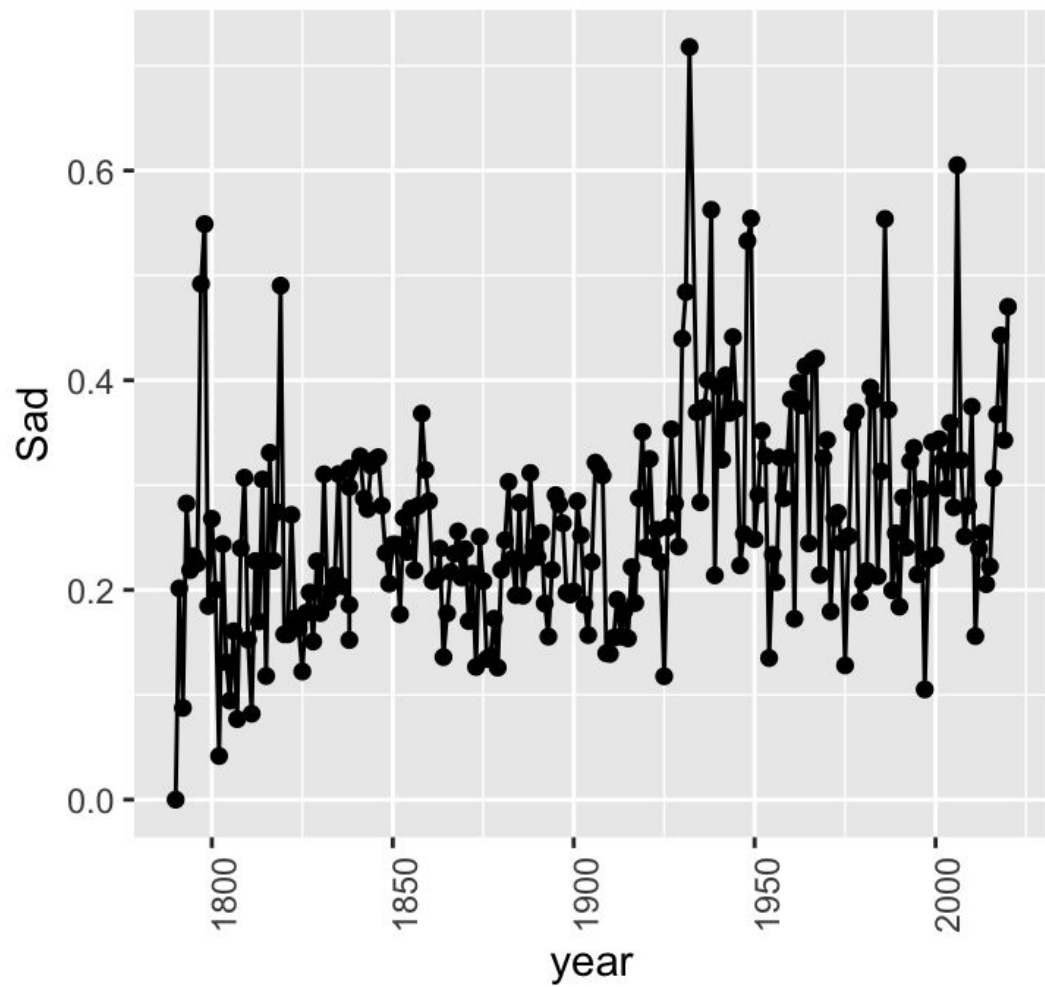












beyond base LIWC: custom dictionaries

What about **moral** language?

Custom dictionaries!

What about **moral** language?
Custom dictionaries!
MAC-D

Heliyon



Volume 10, Issue 6, 30 March 2024, e25940

Research article

Moral universals: A machine-reading analysis of 256 societies

Mark Alfano ^a  , Marc Cheong ^b, Oliver Scott Curry ^c

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Abstract

What is the cross-cultural prevalence of the seven moral values posited by the theory of “morality-as-cooperation”? Previous research, using laborious hand-coding of ethnographic accounts of ethics from 60 societies, found examples of most of the seven morals in most societies, and observed these morals with equal frequency across cultural regions. Here we replicate and extend this analysis by developing a new Morality-as-Cooperation Dictionary (MAC-D) and using Linguistic Inquiry and Word Count (LIWC) to machine-code ethnographic accounts of morality from an

What about **moral** language?
Custom dictionaries!
MAC-D (or MFD if you prefer a dictionary (<https://osf.io/ezn37/>) based on shitty theory)



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Research article

Moral universals: A machine-reading analysis of 256 societies

Mark Alfano ^a ✉, Marc Cheong ^b, Oliver Scott Curry ^c

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The extended Moral Foundations Dictionary (eMFD): Development and applications of a crowd-sourced approach to extracting moral intuitions from text

Published: 14 July 2020
Volume 53, pages 232–246, (2021) [Cite this article](#)

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[Frederic R. Hopp](#), [Jacob T. Fisher](#), [Devin Cornell](#), [Richard Huskey](#) & [René Weber](#) ✉

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Abstract

Moral intuitions are a central motivator in human behavior. Recent work highlights the importance of moral intuitions for understanding a wide range of issues ranging from online radicalization to vaccine hesitancy. Extracting and analyzing moral content in messages, narratives, and other forms of public discourse is a critical step toward understanding how the psychological influence of moral judgments unfolds at a global scale. Extant approaches for extracting moral content are limited in their ability to captu

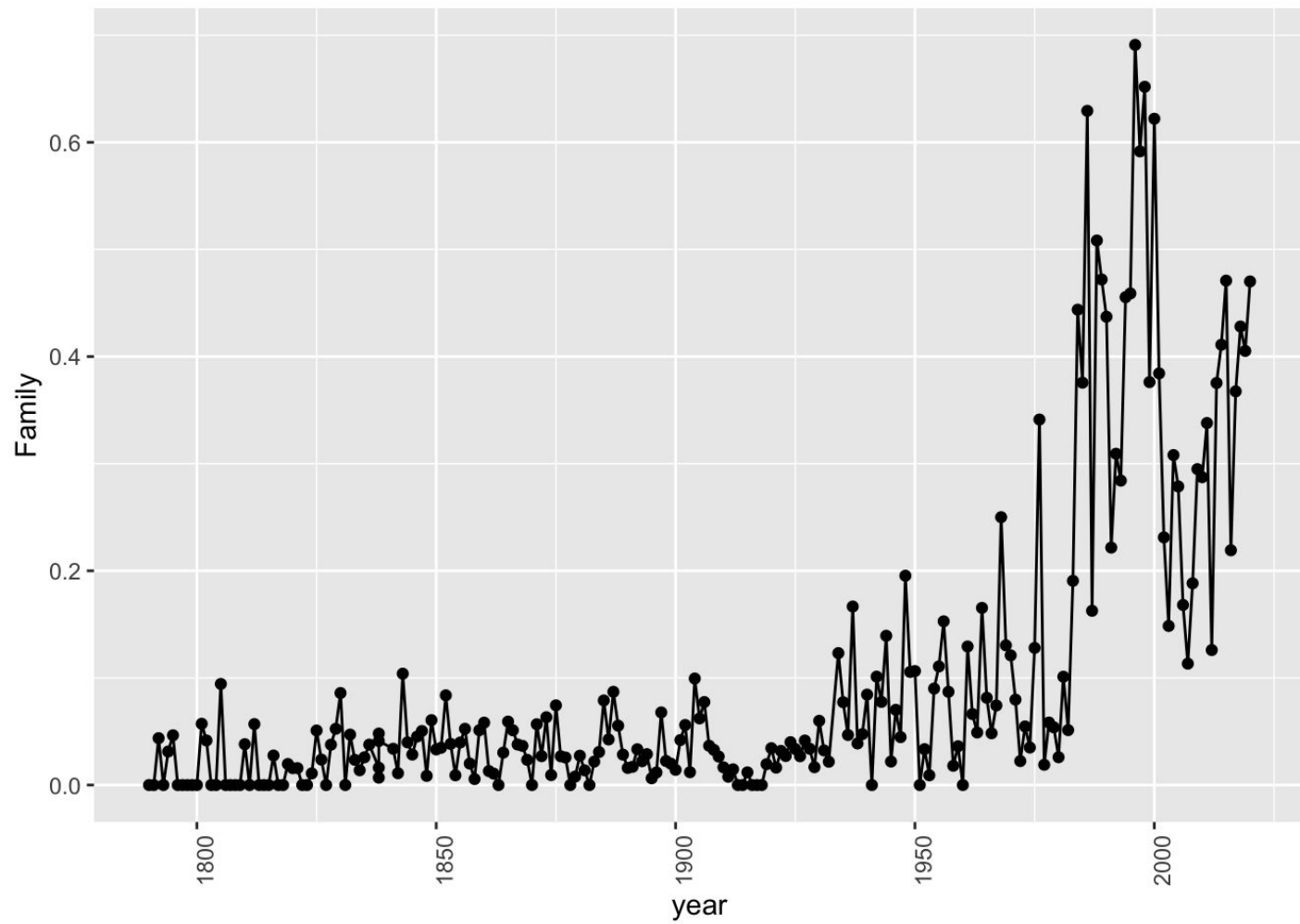


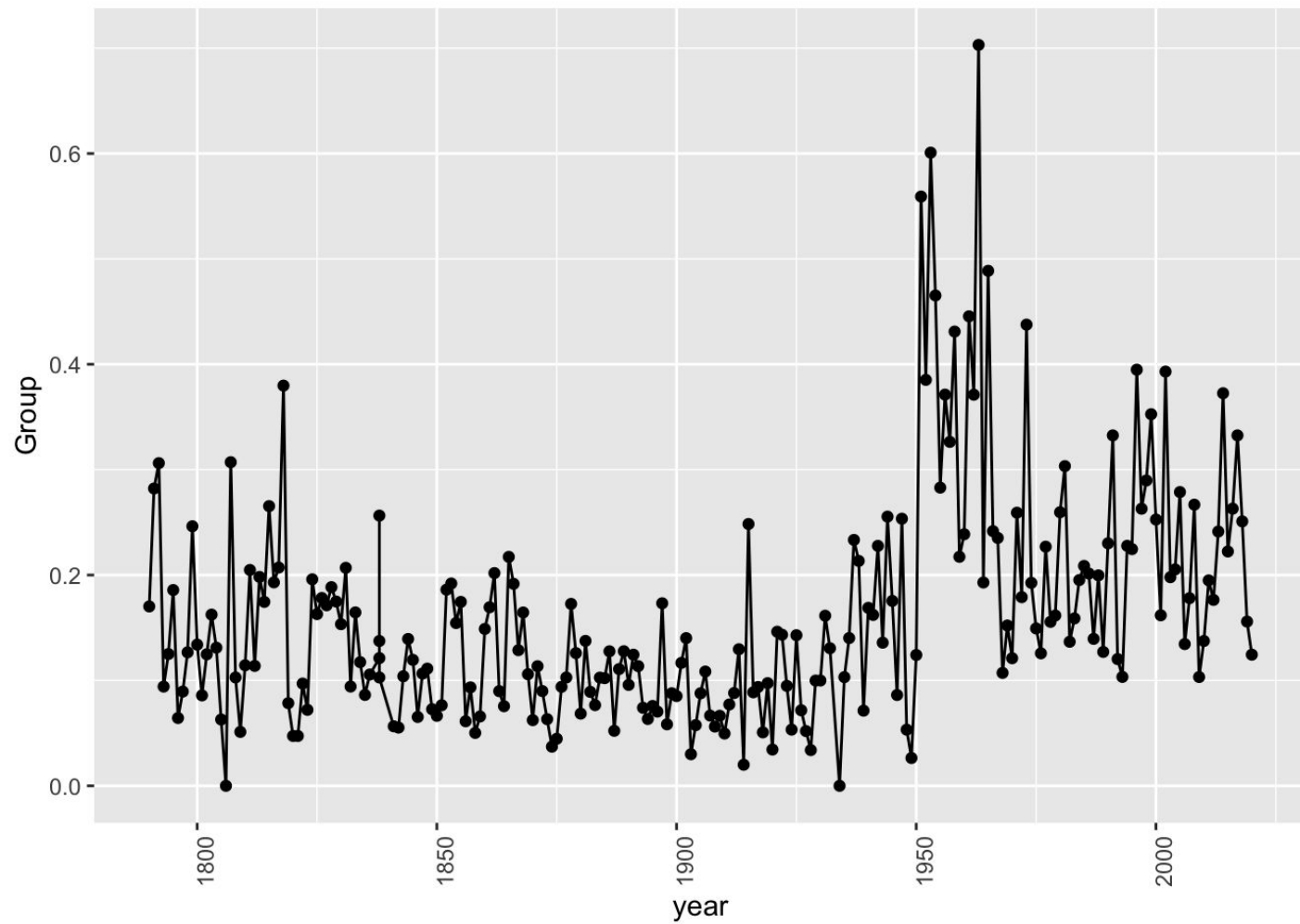
Behavior Research Methods

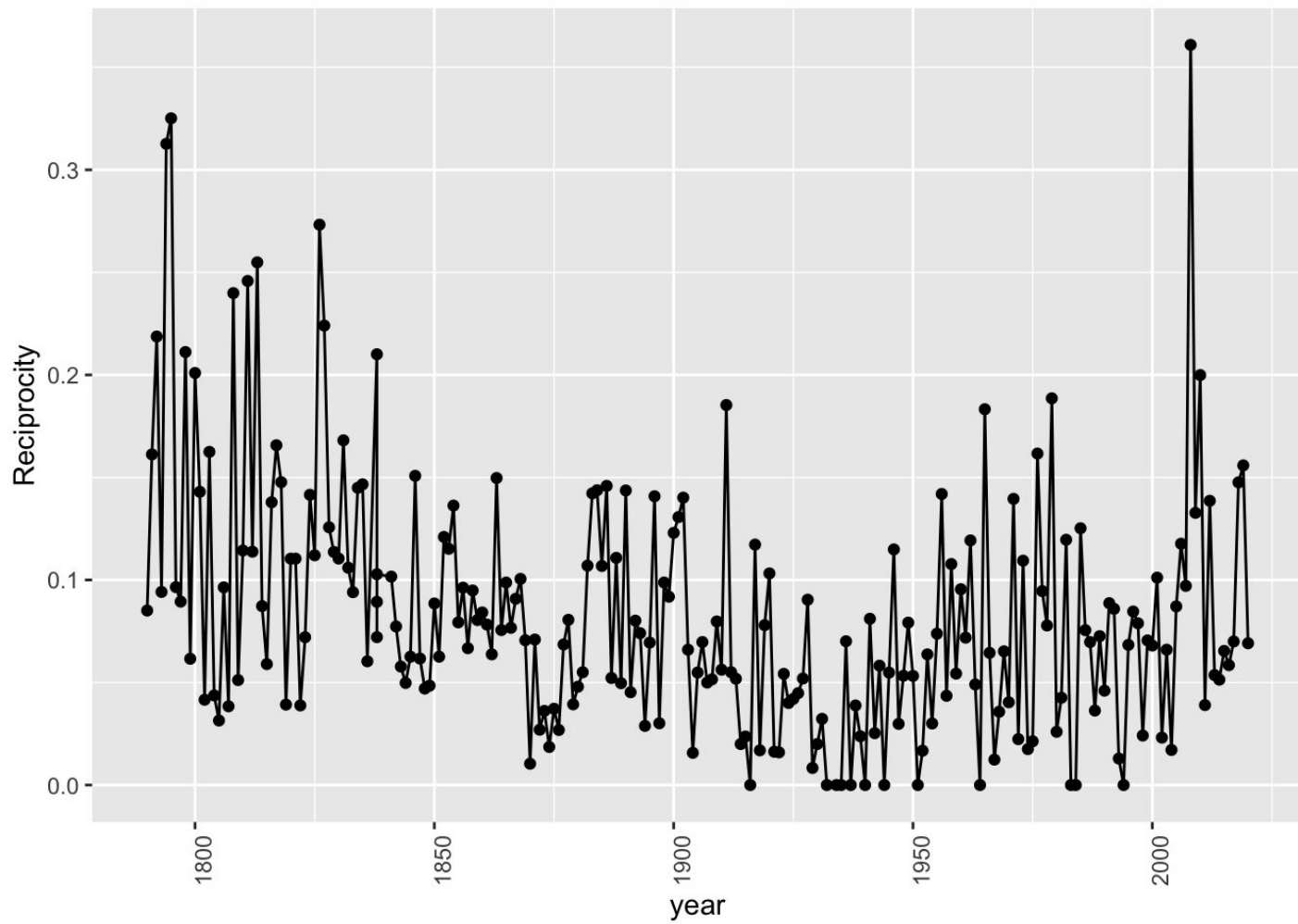
[Aims and scope](#) ➔

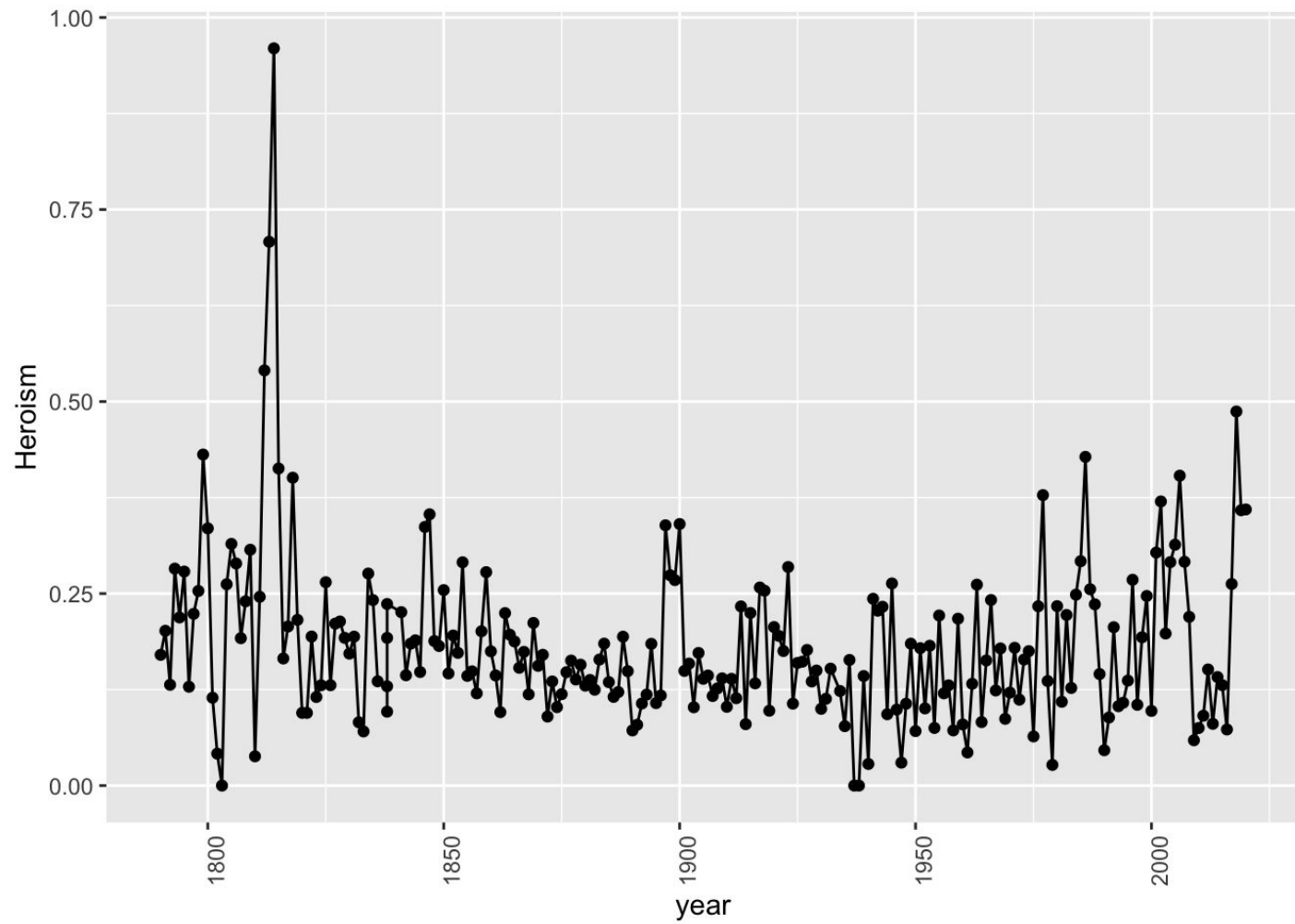
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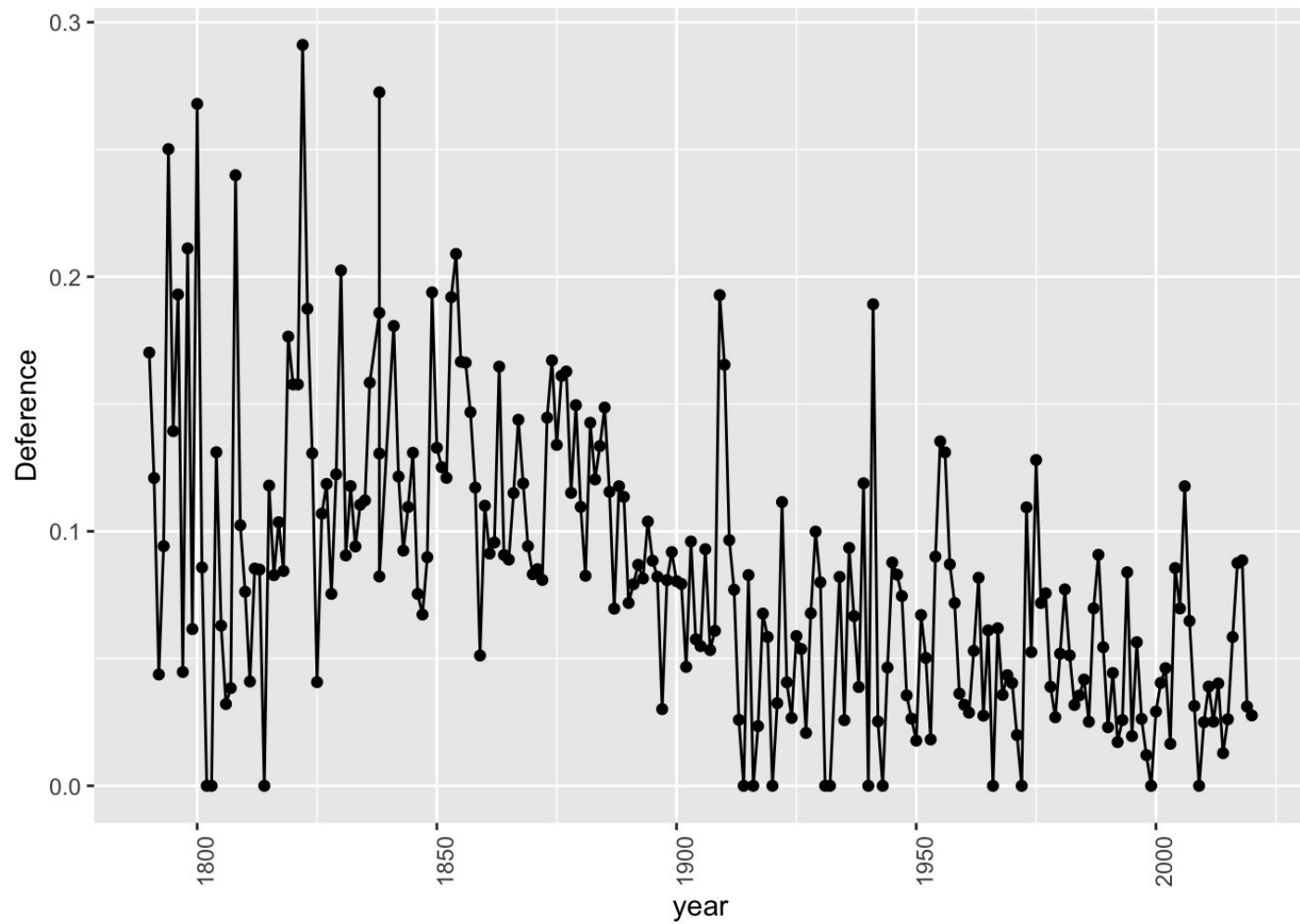


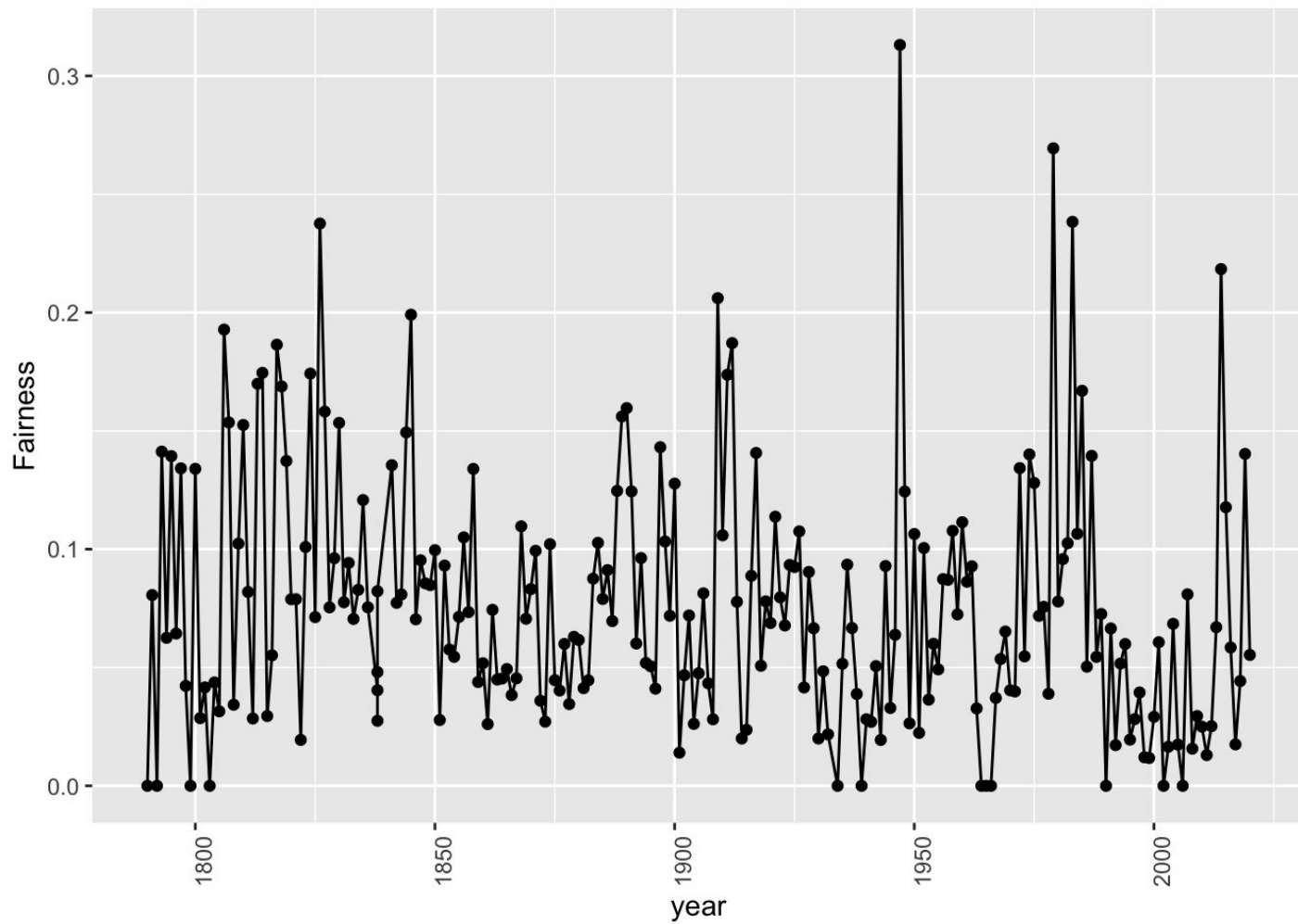


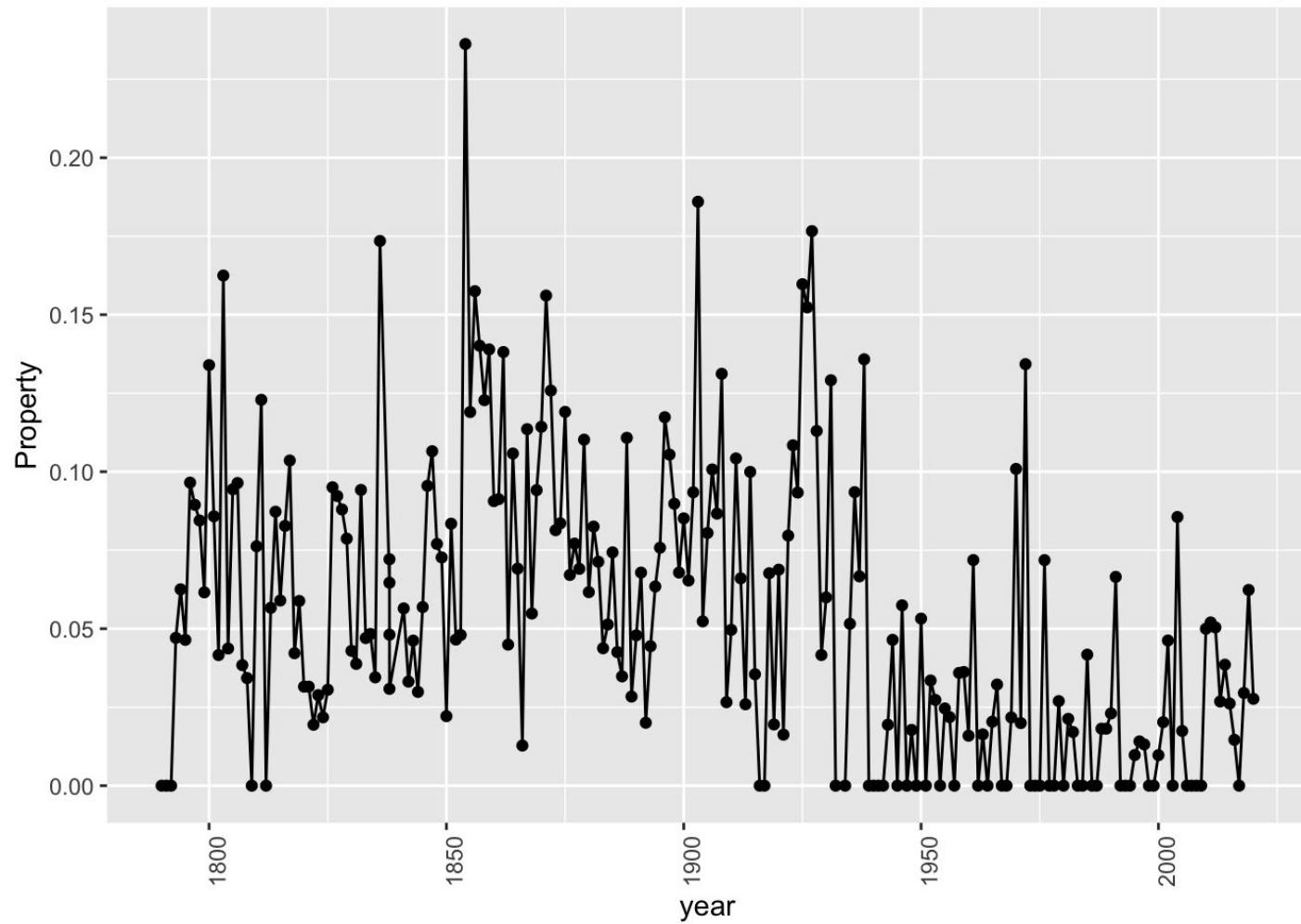












beyond base LIWC: semantic networks



John Rupert Firth

You shall **know a word** by
the **company** that it keeps.

So you need **context**, e.g., **collocations** or **embeddings...**

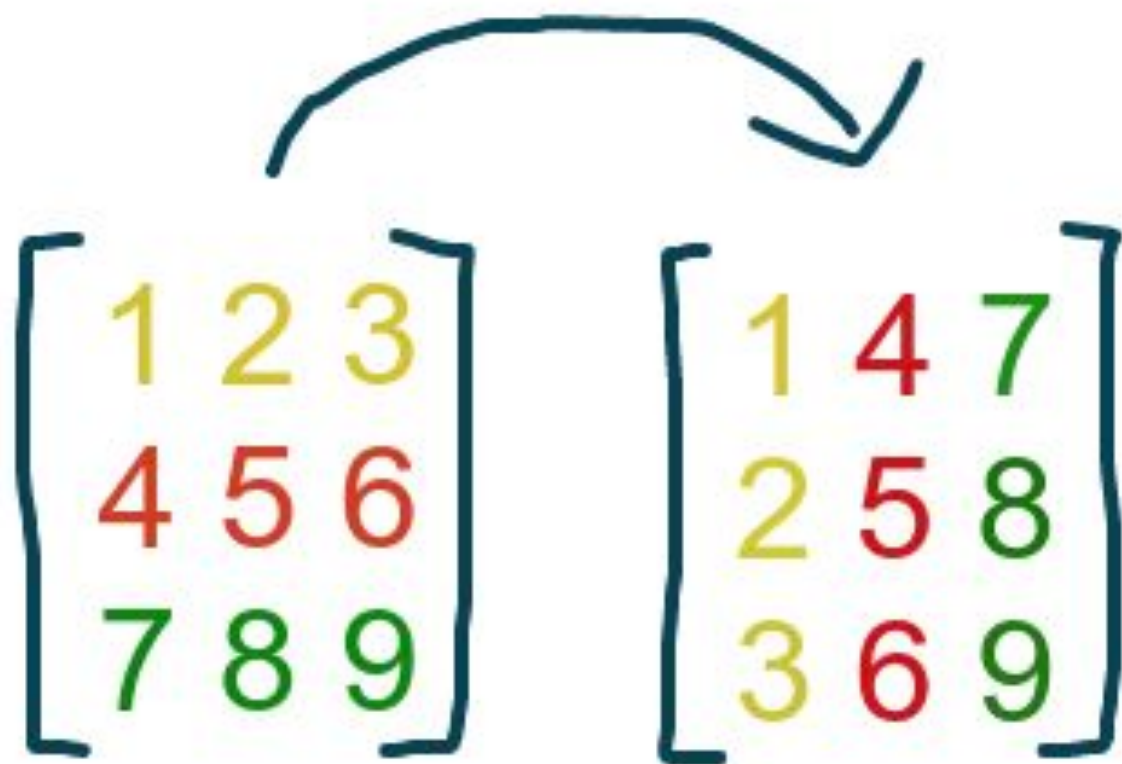


KEEP
CALM
and
RAMSIFY

always has been

it's all just matrix multiplication?

Transpose





Matrix multiplication is **not commutative!**

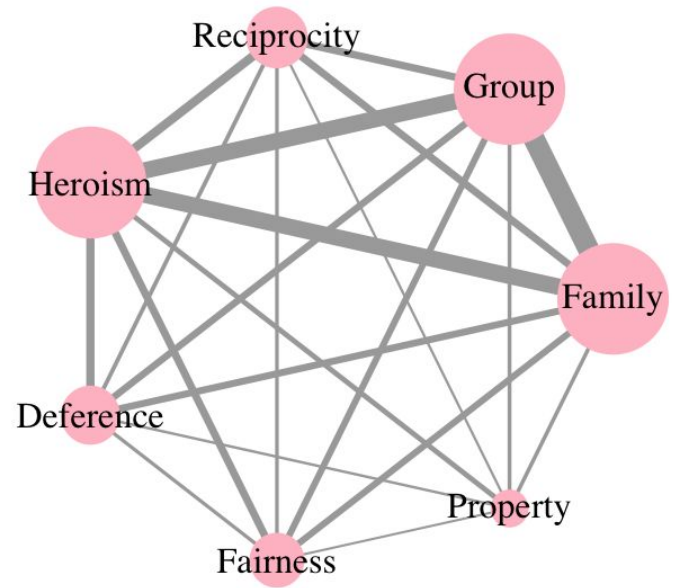
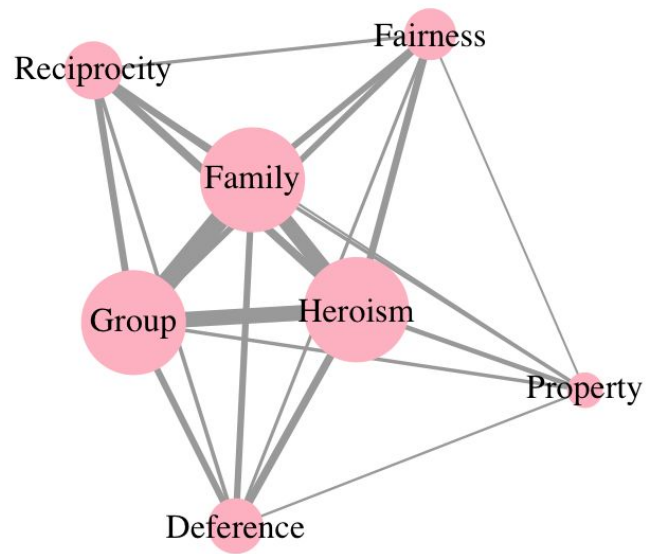
$$M \times T \neq T \times M$$

What question do you want to ask?

Live demo!

(in which Mark takes his life in his hands)

(for more, see <https://tinyurl.com/bdz7chz2>)



Before we end, sticky note feedback

- On the green sticky note, write one thing that went well this session.
- On the red sticky note, write one thing that we can improve on, tomorrow.

Thanks for
your attention!